UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 8 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 8 printed pages

«:> Hak Cipta Universiti Teknologi MARA

QUESTION 1

a) Given $z_1 = -1 - 6i$ and $z_2 = -1 + 3i$, find $z_1 + z_2$ in the form a + bi.

(4 marks)

b) Given $z_1 = -4 + 5i$ and $z_2 = 5 - 3i$, express z_1/z_2 in the form a + bi.

(5 marks)

c) Evaluate i^40 and simplify completely.

(3 marks)

d) Find the modulus and argument of z = 3 + 4i. Give the argument in degrees correct to one decimal place.

(5 marks)

QUESTION 2

a) Solve the simultaneous equations
4x - 2y = 10
x + 3y = -15.

(4 marks)

b) Solve the system of equations simultaneously:
-2x + y - 3z = 4
x - y + 3z = -1
-x - 2y - z = 0

(5 marks)

c) Solve the simultaneous equations
$y = x^2 - 6$
y = 8x - 21.

(5 marks)

C Hak Opta Universiti Teknologi MARA

QUESTION 3

a) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 1x + 2y \leq 6.$

(5 marks)

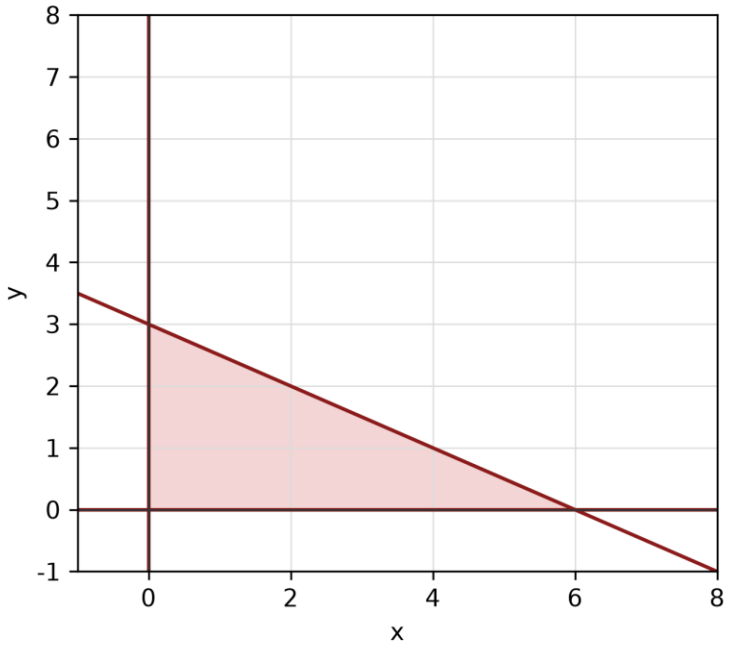


Figure: feasible region.

b) Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (3, 0) and (0, 8).

(5 marks)

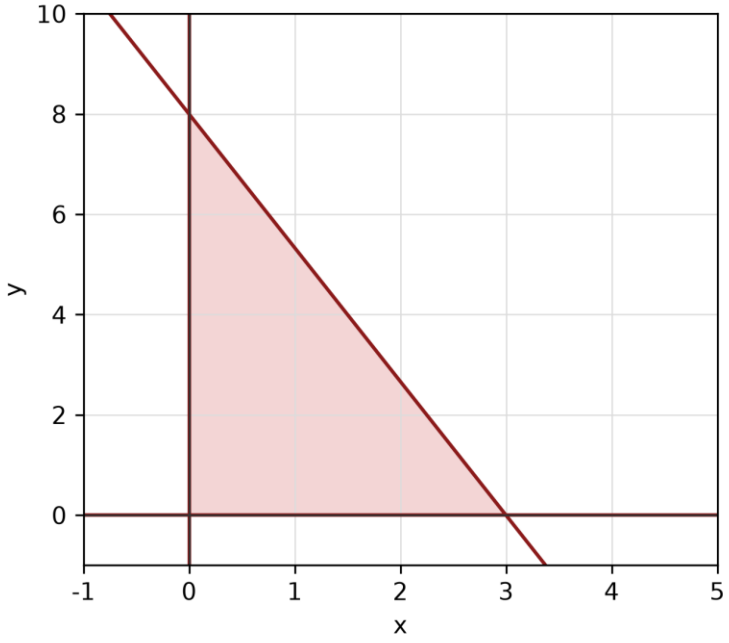


Figure: shaded triangular region.

QUESTION 4

a) Use long division to find the quotient and remainder when $f(x) = -4x^3 + 4x^2 - 3x + 2$ is

divided by x - 1.

(5 marks)

b) Use synthetic division to divide f(x) = x^3 + 5x² - x + 5 by x + 3. State the quotient and remainder.

(5 marks)

c) Show that x - 4 is a factor of f(x) = 3x^3 - 9x² - 17x + 20 by using the factor theorem. Hence factorise f(x) completely.

(5 marks)

d) Use the remainder theorem to find the remainder when f(x) = 2x^3 + 2x² - 5x - 2 is divided by x + 3.

(3 marks)

e) Factorise f(x) = 3x^3 + 6x² - 27x - 54 completely.

(5 marks)

QUESTION 5

a) Solve the exponential equation 3^(4 - 3x) = 3^(-x - 2).

(5 marks)

b) Solve the logarithmic equation log_2(x) + log_2(x - 4) = 5.

(5 marks)

c) Express log_5(5) + log_5(5) as a single logarithm and evaluate it.

(3 marks)

QUESTION 6

a) Solve the trigonometric equation 2cos² x - 2cos x = 0 for 0° ≤ x < 360°.

(5 marks)

b) Prove the identity (1 + cos x)/sin x + sin x/(1 + cos x) = 2 csc x.

(5 marks)

c) Given sin A = 3/5 and A is in quadrant 1, find the exact values of cos A, tan A and sec A.

(4 marks)

d) For y = -3 sin(3x + 0) + -2, state the amplitude, period and phase shift, then sketch the graph for -180° ≤ x ≤ 180°.

(5 marks)

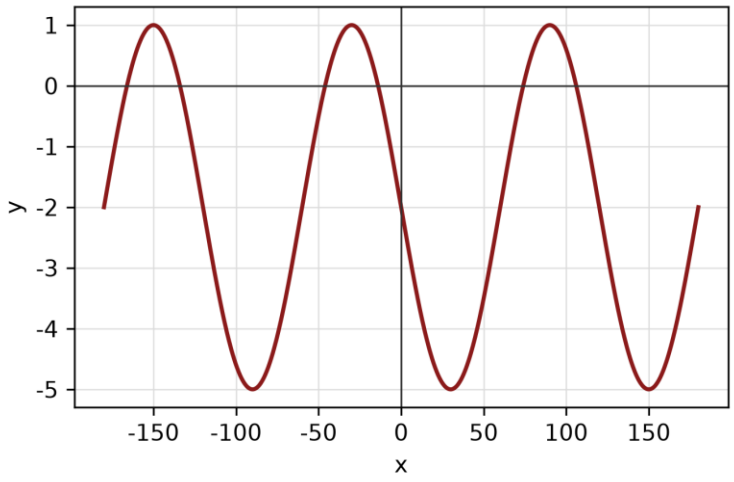


Figure: trigonometric graph.

e) Given sin A = 3/5 where A is in quadrant II and cos B = 4/5 where B is in quadrant IV, find sin 2A and cos(A + B).

(5 marks)

QUESTION 7

a) A circle has radius 9 cm. Find the length of the arc subtended by a central angle of 30°.

(4 marks)

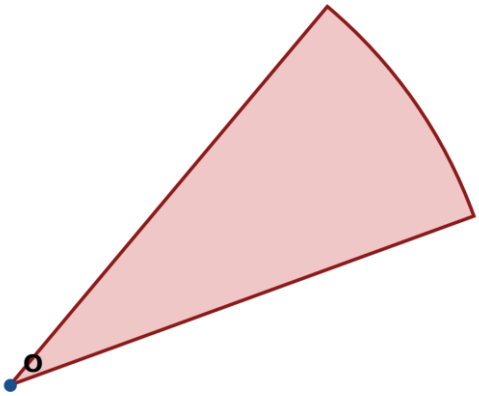


Figure: arc and sector.

b) Find the area of the sector of a circle with radius 5 cm and central angle 135°.

(4 marks)

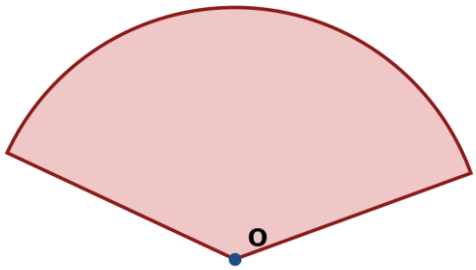


Figure: sector.

c) Find the perimeter of the sector with radius 12 cm and central angle 30°.

(4 marks)

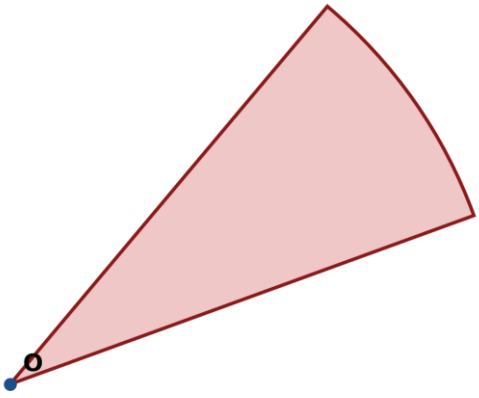


Figure: sector.

d) Convert 45° to radians, and convert π radians to degrees.

(3 marks)

«:> Hak Cipta Universiti Teknologi MARA

QUESTION 8

a) Solve the triangle with sides a = 8 cm, b = 15 cm and c = 17 cm. Find all three angles.

(5 marks)

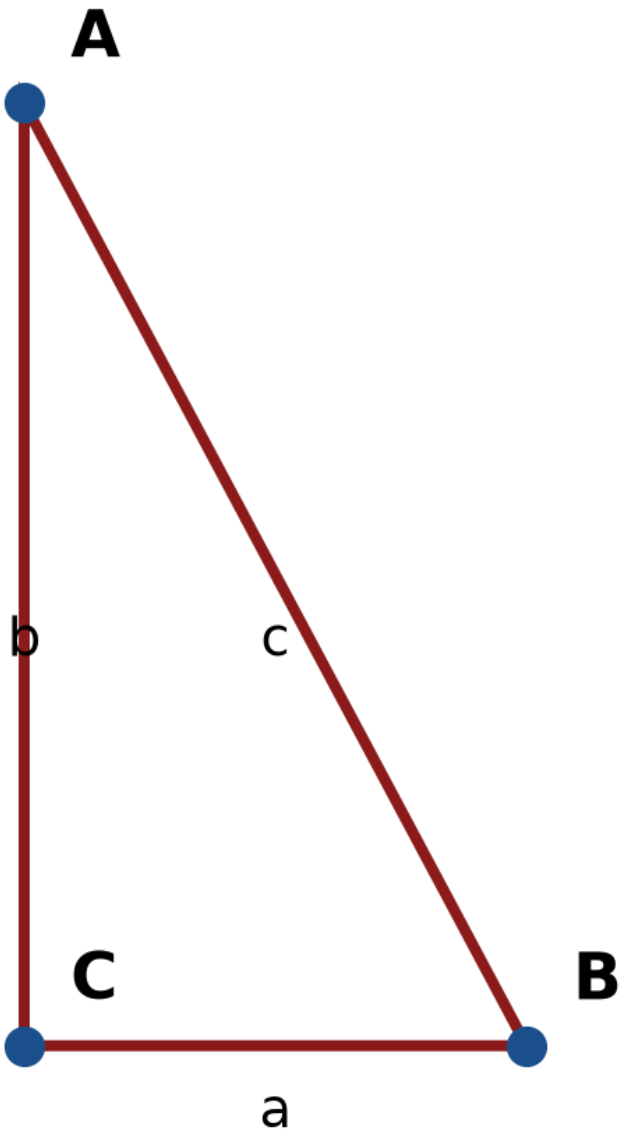


Figure: right triangle.

b) In triangle ABC, A = 30°, B = 45° and a = 11 cm. Use the sine rule to find b.

(4 marks)

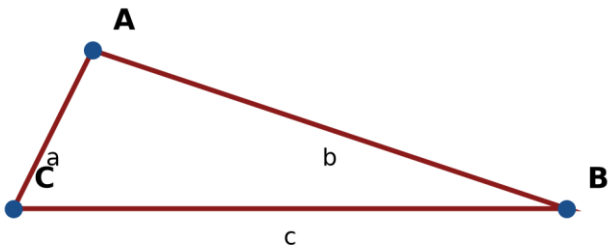


Figure: triangle ABC.

c) In triangle ABC, b = 5 cm, c = 6 cm and A = 120°. Use the cosine rule to find a.

(4 marks)

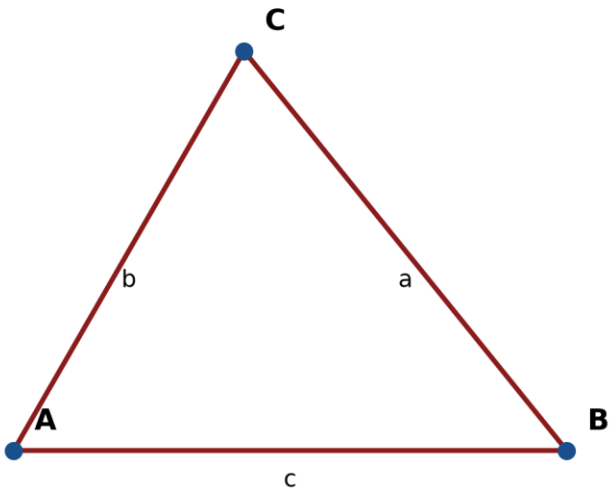


Figure: triangle ABC.

d) Find the area of triangle ABC with a = 15 cm, b = 6 cm and included angle C = 90°.

(4 marks)

e) A right triangle has legs 10 cm and 9 cm. A sector of radius 4 cm and angle 45° is cut from it. Find the area of the remaining shaded region.

(5 marks)

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

C Hak Opta Universiti Teknologi MARA